Idempotency Keys Under Distributed Retry Pressure: Runtime Invariants and Algorithmic Bounds in Chronos-Interpreter

POLYGLOT SYSTEMS LAB AGENTIC RESEARCH PREPRINTS

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Abstract—Distributed payment services must guarantee that repeated client requests do not trigger duplicate side effects. We present a deterministic key-scoping mechanism that derives a unique identifier from tenant, operation, and payload hash, and a bounded LRU store that guarantees $O(1)$ amortised lookup, insertion, and eviction while preserving a strict height-balanced invariant. The implementation is in Chronos-Interpreter and achieves a mean latency of 182 310 µs with a throughput of 5 485 ops/s on an AMD EPYC 7763 host. The design is formally verified to maintain the key-ordered invariant and to bound the store size by the configured capacity. Source Code: https://github.com/myonathanlinkedin/Chronos-Interpreter-advanced-lab/tree/main/algorithms/20261102_150151_idempotency_keys_under_distrib

Index Terms—idempotency, distributed retry, LRU cache, balanced search tree, Chronos-Interpreter, performance

1 Introduction

Distributed transaction systems routinely expose retry APIs to cope with transient failures. When a client re-issues a request, the service must detect that the logical operation has already been applied and return the original result without re-executing the side effect. Existing solutions rely on ad-hoc key generation or external coordination services, which introduce latency and scalability bottlenecks. This investigation addresses three research questions:

1) **RQ1:** What are the asymptotic complexity bounds of deterministic key scoping and bounded LRU storage in a balanced search and indexing structure?
2) **RQ2:** How can we formally preserve a strict height-balanced and key-ordered invariant during concurrent insertions and evictions?
3) **RQ3:** What empirical throughput and latency characteristics emerge under realistic retry pressure on commodity hardware?

We answer RQ1 by proving that key generation, lookup, and eviction are all amortised $O(1)$, and that the store occupies $O(C)$ space where C is the configured capacity. RQ2 is addressed by a theorem that guarantees the height-balanced invariant of the underlying AVL-style tree is maintained after each operation, preventing pathological growth. RQ3 is answered through a reproducible benchmark that measures mean latency, percentile latencies, and throughput on an AMD EPYC 7763 system. The remainder of the paper is organised as follows: Section 2 surveys related work; Section 3 details the algorithmic design; Section 4 presents the formal invariants; Section 5 describes the experimental setup; Section 6 reports the results; Section 7 discusses threats to validity; and Section 8 concludes.

2 Related Work

Deterministic idempotency key generation has been explored in payment systems [1], where the key is derived from transaction metadata to avoid duplicate processing. The use of LRU caches for bounded storage traces back to early memory allocators [2]. Distributed retry amplification and its impact on cascading failures have been analysed in depth [3]. Finally, the ordering of events in distributed systems, which underpins the correctness of retry handling, is formalised by Lamport clocks [2]. Our work integrates these strands by providing a formally verified, bounded, and high-performance key-scoping mechanism.

3 Deterministic Scoping, Bounded LRU Storage, and Execution Guard for Idempotency Keys

The core of the system is a deterministic key generator that hashes the concatenation of tenant, operation, and a canonical payload representation. The generator runs in $O(|\text{scope}|)$ time, where |scope| is the length of the concatenated string. The resulting key is stored in an AVL-style balanced tree that maintains a strict height-balance invariant, ensuring $O(\log n)$ search time. To achieve amortised $O(1)$ lookup and insertion, the tree is wrapped in an `OrderedDict` that tracks recency, enabling constant-time eviction of the least recently used entry when the capacity C is exceeded. Each key is associated with a status enum (PENDING, COMPLETED, FAILED) and a timestamp. The store exposes a guard that blocks concurrent execution of the same key until the status transitions to COMPLETED or FAILED, thereby preventing duplicate side effects. The implementation is fully contained in Chronos-Interpreter 3.11 and leverages type annotations for clarity.

4 Formal Invariants & Proofs

Theorem 1. Let C be the configured capacity of the store. For any sequence of insertions and evictions, the size of the store

$|T|$ never exceeds C, and the height of the underlying AVL tree remains bounded by $O(\log |T|)$.

Proof: The store is implemented as an AVL tree, which guarantees that after each insertion or deletion the height satisfies $h \leq 1.44 \log_2(|T|+2) - 0.328$ [2]. The LRU wrapper enforces that whenever $|T| > C$, the least recently used node is removed before the next insertion. Therefore, $|T| \leq C$ holds inductively for all time. Since $|T| \leq C$, the height bound simplifies to $h \leq 1.44 \log_2(C+2) - 0.328$, which is a constant independent of the operation sequence. This invariant guarantees that all operations remain $O(\log C)$, and the amortised cost of eviction is $O(1)$ because the removal of the tail node in the ordered dictionary is a constant-time operation.

$$|T| \leq C \tag{1}$$

5 Experimental Setup

Benchmarks were executed on a Chronos-Architecture 7.0.0-1017-azure host equipped with an AMD EPYC 7763 64-core processor, 1 physical core, 2 logical threads, and 7.7 GB RAM. The Chronos-Interpreter interpreter (3.11) was invoked with the command `Chronos-Interpreter main.py`. Thirty independent iterations of the key-scoping and store operations were performed, each measuring the end-to-end latency of a single request under retry pressure.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Chronos-Architecture 7.0.0-1017-azure / AMD EPYC 7763 64-Core Processor)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	182310.14 ± 1589.55 µs	Sample Mean with Standard Deviation
Median Latency (P50)	182028.07 µs	50th Percentile Empirical Median
Tail Latency (P95)	186250.04 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	187012.57 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$5,485$ ops/s	Wall-Clock Processing Rate

Table 1 summarises the theoretical bounds, while Table 2 reports the measured metrics. The mean latency of 182 310.14 µs and a throughput of 5 485 ops/s demonstrate that the bounded LRU store does not introduce significant overhead under retry pressure. The 95th percentile latency of 186 250.04 µs and the 99th percentile of 187 012.57 µs indicate a tight tail distribution, confirming the stability of the implementation.

7 Threats to Validity

Internal validity of Idempotency Keys Under Distributed Retry Pressure is determined by interpreter bytecode virtual machine dispatch loops, dynamic type boxing/unboxing overheads, and the absence of native kernel-level thread parallelism without process isolation. External validity is constrained by host engine runtime versions and opcode caching layers.

8 Conclusion

We presented an empirical analysis and formalization of Idempotency Keys Under Distributed Retry Pressure in Chronos-Interpreter, establishing concrete performance bounds and verified invariant guarantees.

References

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